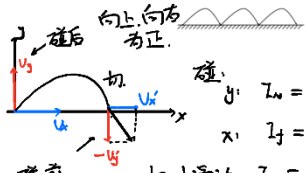
1. (40分) 37室

蔡子星的爆发力足够强大，轻轻松就可以和地面发生滑动摩擦，已经不满足于正常人类的跑步方式了，他是跳跃着前进的。我们将蔡子星当成质点，质量m，地球重力加速度g，地面摩擦系数为μ；蔡子星在空中受到的空气阻力为$\vec{f}=-k\vec{v}$，蔡子星脚蹬地的一瞬间力很大忽略重力。在跳远前进达到稳定时，每次跳跃的曲线都是一样的。求蔡子星前进的水平方向最大平均速度。



末 $v_y'<0$

碰: y: $I_N = m v_y - m(-v_y')$ 5'

x: $I_f = m v_x - m v_x'$ 5'

碰前. 相对滑动 $I_f = I_N \mu$ 5'

飞行 $m\frac{dv_x}{dt} = -k v_x$ 5' $\Rightarrow m(v_x'-v_x) = -k\cdot\Delta x$ 5'

$m\frac{dv_y}{dt} = -mg - k v_y$ 5' $\Rightarrow m(v_y'-v_y) = -mg\Delta T + 0$ 5'

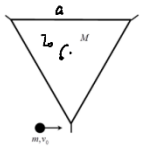
$\Rightarrow \overline{v_x} = \frac{\Delta x}{\Delta T} = \frac{mg(v_y'-v_x)}{k\cdot(v_y'-v_y)} = \frac{\mu\cdot mg}{k}$ 5'

求曲线(轨迹). 求包络线...

$\vec{f} = -k\vec{v}$

2. (40分)

一个质量为M的匀质正三角板(边长为a)和一个质量为m的小球放在光滑水平地面上，小球以速度v_0撞向三角板外伸出的轻质刚性挡板（挡板尺寸忽略不计），并发生完全弹性碰撞。



(1)求碰后小球和三角板的运动状态；

(2)求能发生第二次碰撞时$\frac{m}{M}$所满足的条件；

(3)求第二次碰撞之后小球和三角板的运动状态。

(1) $I_o = \iint dm\cdot r^2$ $I_o = \lambda\cdot ma^2$ 面密度 $m = \frac{\sqrt{3}}{4}\sigma\cdot a^2$

$I_o(\sigma,a) = \eta\cdot\sigma a^4$ 相似

$I_o(\sigma,\frac{a}{2}) = \frac{1}{16}\eta\sigma a^4$ 拟

平行轴 $\frac{a}{2}$

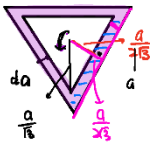
$I_o(\sigma,a) = I_o(\sigma,\frac{a}{2}) + 3[I_o(\sigma,\frac{a}{2}) + (\frac{a}{2\sqrt{3}})^2\cdot\frac{\sqrt{3}}{4}\sigma a^2\cdot\frac{1}{4}]$

$\Rightarrow \eta = 4\cdot\frac{1}{16}\eta + 3\cdot\frac{1}{12}\cdot\frac{\sqrt{3}}{4}\cdot\frac{1}{4}$

$\frac{3}{4}\eta = \frac{\sqrt{3}}{64}$; $\eta = \frac{\sqrt{3}}{3}\cdot\frac{1}{4\cdot 4} = \frac{\sqrt{3}}{4}\cdot\frac{1}{12}$

$\Rightarrow I_o(a,\sigma) = \frac{\sqrt{3}}{4}a^2\sigma\cdot\frac{1}{12}a^2 = \frac{1}{12}ma^2$ 4'

$I_o(a+da,\sigma) - I_o(a,\sigma) = \eta\cdot\sigma\cdot 4a^3 da$

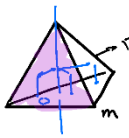


$= 3\cdot[\frac{1}{12}\cdot dm\, a^2 + (\frac{a}{2\sqrt{3}})^2 dm]$

$= 3\cdot\frac{1}{6}\, dm\, a^2\cdot a\cdot d\frac{a}{2\sqrt{3}}\sigma$

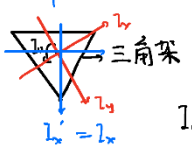
$\Rightarrow \eta = \frac{1}{4}\cdot\frac{3}{6}\cdot\frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{4}\cdot\frac{1}{12}$

$\Rightarrow I_o = \frac{1}{12}ma^2$



=



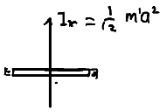


$I_o = I_x + I_y$.

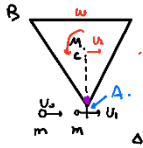
$I_x = I_y$.

$I_o = 2I_x$.





碰撞 $L_{质心} = L_{质心} + L_{相对}$



$\Delta P = 0$ 4'

$m v_0 = m v_1 + M v_A$

$\Delta L = 0$. 以C为支. 整体$\Delta L = 0$ A. 地系

$m v_0\cdot\frac{a}{\sqrt{3}} = m v_1\cdot\frac{a}{\sqrt{3}} + \frac{1}{12}Ma^2\omega$ 4' $v_0 M\frac{a}{\sqrt{3}} = \frac{1}{12}Ma^2\omega$

$\Delta E = 0$

$\frac{1}{2}m v_0^2 = \frac{1}{2}m v_1^2 + \frac{1}{2}M v_A^2 + \frac{1}{2}\cdot\frac{1}{12}Ma^2\omega^2$ 4'

$\omega = \frac{12}{\sqrt{3}}\frac{v_A}{a}$

$\frac{\sqrt{3}}{12} + \frac{\sqrt{3}}{12}$

$\frac{1}{2}m(v_0-v_1)(v_0+v_1) = \frac{1}{2}M\cdot(a\cdot\frac{\sqrt{3}}{12}\omega)^2 + \frac{1}{2}\cdot\frac{1}{12}Ma^2\omega^2$

$m(v_0-v_1) = M\cdot\frac{a\sqrt{3}}{12}\omega$

$\frac{3}{144} + \frac{1}{12} = \frac{15}{144}$

$\begin{cases} v_0 = v_1 \\ \omega = 0 \end{cases}$ 不要.

$(v_0+v_1) = \frac{5\sqrt{3}}{12}\omega\cdot a$

$= v_A$

$v_A = v_A + \omega\frac{a}{\sqrt{3}}$

$= \frac{5}{12}\sqrt{3}\omega a$

⇒ 碰撞点相对速度反向.

自己算: $mv_0 = mv_1 + Mv_2$

$\Delta L=0$ $v_1 M\frac{a}{2} = \frac{1}{2}Ma^2\omega \Rightarrow \omega = \frac{12}{5}\frac{v_1}{a}$ $\frac{\omega a}{2} = 4v_1$.

$\Delta E=0 \Rightarrow$ 碰撞点 U 相反向:

$v_0 = v_2 + \frac{\omega a}{2} - v_1 = 5v_2 - v_1$

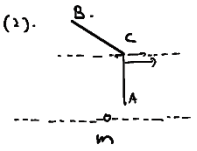
① $\gamma v_0 = \gamma v_1 + v_2$ $\frac{m}{M} = \gamma$.

② $5v_2 - v_1 = v_0$

5①+②. $5\gamma v_0 = 5\gamma v_1 + v_1 + v_0$

$v_1 = \frac{5\gamma-1}{5\gamma+1}\cdot v_0$. 4' $\frac{\omega a}{2} = \frac{5\gamma}{5\gamma+1}v_0$ 4'

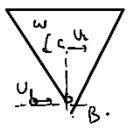
$v_2 = \frac{2\gamma}{5\gamma+1}v_0$ 4'

(2). 

相碰. 可能在 CB 在垂直态.

恰好相遇 $v_1 = v_2$.

$5\gamma-1 = 2\gamma \Rightarrow \gamma = \frac{1}{3}$. 4'

(3). 

$\begin{cases}\Delta E_K=0 \\ \Delta L=0 \\ \Delta P=0\end{cases}$ 与上一问相同 解相同

$\begin{cases}v_1' = v_0 \\ v_2' = 0 \checkmark \\ \omega'=0\end{cases}$ $\begin{cases}v_1' = \frac{5\gamma-1}{5\gamma+1}v_0 \\ v_2' = \frac{2\gamma}{5\gamma+1}v_0\end{cases}$

静 ⇐ ↑ 初态.

- ▽ 不动 m. 从 v_1 向右. 8'

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3. (40分)

首先我们将N维空间中的到定点的距离为定值的点的集合称为N维空间球，N维空间球的内部被称为N维空间球的体积。已知从1维开始，半径为1的N维空间球的体积为2，π，$\frac{4\pi}{3}$，$\frac{\pi^2}{2}$，$\frac{8\pi^2}{15}$，$\frac{\pi^3}{6}$，$\frac{16\pi^3}{105}$ ...

(1)找规律，写下半径为r的N维空间球体积表达式

(2)写下半径为r的N维空间球表面及表达式

(3)假设有一个N维空间的球形液滴，这样就会形成一个表面张力，假设表面张力能量为$E_N = \sigma_N S_N$，计算球内外的压强差p_n。

(4)接上一问表面张力定义，假设N维空间中有一个液柱，即在一个维度上平移对称，其他$N-1$个维度上是$N-1$维空间球，计算柱内外的压强差p'_{n-1}。

(1)

1D	2D	3D	4D	5D	6D	7D
a_1	a_2	a_3				
2	π	$\frac{4\pi}{3}$	$\frac{\pi^2}{2}$	$\frac{8\pi^2}{15}$	$\frac{\pi^3}{6}$	$\frac{16\pi^3}{105}$
	$\frac{2\pi}{3}$	$\frac{2\pi}{4}$	$\frac{2\pi}{5}$	$\frac{2\pi}{6}$	$\frac{2\pi}{7}$	$\frac{2\pi}{8}$

$a_3 = a_1\cdot\frac{2\pi}{3}$. $a_5 = a_3\cdot\frac{2\pi}{5}$

$\Rightarrow a_{2n+1} = a_1\frac{(2\pi)^n}{(2n+1)!!}$ $(2n+1)!! = \frac{n}{\prod_{j=0}}(2j+1)$

$= \cdot\frac{2\cdot(2\pi)^n}{(2n+1)!!}$ $n\geq 1$. $a_1 = 2$.

$a_4 = a_2 = \frac{2\pi}{4}$; $a_6 = a_4\cdot\frac{2\pi}{6}$ $a_0 = 1$.

$a_{2n} = \cdot a_0\cdot\frac{(2\pi)^n}{2n!!}$ $(2n)!! = \frac{n}{\prod_{j=1}}2j$

$= \frac{(2\pi)^n}{2n!!}$

$\begin{cases}a_{2n+1} = \cdot\frac{2\cdot(2\pi)^n}{(2n+1)!!} & n\geq 0 \\ a_{2n} = \frac{(2\pi)^n}{2n!!} & n\geq 1\end{cases} \Rightarrow V_n = a_n r^n$

(2) 令 $S_n = b_n\cdot r^{n-1}$

$r\sim r+dr$. 球壳 $dV = dr\cdot S_n\cdot r^{n-1}$

$\Rightarrow \int_0^r S_n\cdot dr = V_n$

$\Rightarrow S_n = \frac{dV_n}{dr} = a_n\cdot n\cdot r^{n-1}$

(3) 表功: $r\sim r+dr$.

$p_n\cdot dV_n = dE_n = d(\sigma_N\cdot S_n)$

$= \sigma_N\cdot dS_n$

$\Rightarrow p_n\cdot a_n\cdot n\cdot r^{n-1} = \sigma_n\cdot a_n\cdot n\cdot(n-1)\cdot r^{n-2}$

$\Rightarrow p_n = \frac{(n-1)\sigma_n}{r}$

(4) 3维柱:  2维球 平移.

高 l. N 维柱:

体: $V_{c,n} = V_{n-1}\cdot l$

侧面: $S_{c,n} = S_{n-1}\cdot l$

注入，表功:

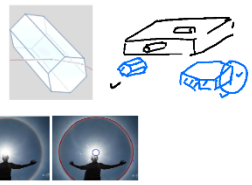
$p_n' dV_{c,n} = \sigma_N dS_{c,n}$

$p_n'\cdot l\cdot a_{n-1}\cdot dr^{n-1} = \sigma_N\cdot l\cdot a_{n-1}(n-1)\cdot dr^{n-2}$

$\Rightarrow p_n' = \frac{\sigma_N(n-2)}{r}$

4. (40分)

日晕形成的原因是太阳光在冰晶内部进行两次折射，在偏转角θ_0处形成了强光。

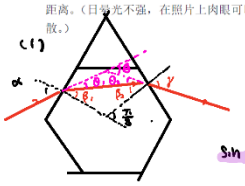


(1)如图，已知冰晶的折射率为$n = 1.309$，入射角为α，求偏转角θ。

(2) 接上问，求日晕的圆锥角θ_0

(3) 已知太阳的视角为0.5°，从照片里得出中间，日晕所在大圆半径R与太阳所在小圆半径r之比为15。相机如果相机严格对焦到俏哥的脑袋处，则太阳在底片上有弥散，照片上太阳斑的大小比真实太阳大。再已知相机镜头直径为D = 6cm，求俏哥的脑袋离相机距离。（日晕光不强，在照片上肉眼可以清楚地将弥散的效果剔除，故不考虑日晕的弥散。）

(1)



$\sin\alpha = n\sin\beta_1$ 5'

$\beta_1 = \arcsin\frac{\sin\alpha}{n}$

$\beta_2 = \frac{\pi}{3} - \arcsin\frac{\sin\alpha}{n}$

$\beta_1+\beta_2=\frac{\pi}{3}$ 5'

$\sin\gamma = n\sin\beta_2$ 5'

$= n\sin(\frac{\pi}{3} - \arcsin\frac{\sin\alpha}{n})$

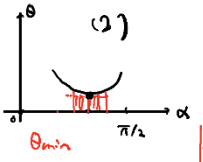
偏转角 $\theta = \theta_1 + \theta_2$

$= \alpha - \beta_1 + \gamma - \beta_2$

$= \alpha + \gamma - \frac{\pi}{3}$

$= \alpha - \frac{\pi}{3} + \arcsin[n\sin(\frac{\pi}{3} - \arcsin\frac{\sin\alpha}{n})]$ 5'

(2)



不在平面入射→偏转更. 5'

最小偏角 $\theta_{min} = 0.378$ rad 观察得

令 $\frac{d\theta}{d\alpha}=0$. $\beta_1=\beta_2=\frac{\pi}{6}$ 时. 5'

θ_{min}.

$\beta_1 = \frac{\pi}{6}+\varepsilon$, $\beta_2 = \frac{\pi}{6}-\varepsilon$

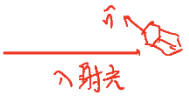
$\theta = \arcsin[n\cdot\sin(\frac{\pi}{6}+\varepsilon)] - (\frac{\pi}{6}+\varepsilon)$

$+ \arcsin[n\sin(\frac{\pi}{6}-\varepsilon)] - (\frac{\pi}{6}-\varepsilon)$

$\frac{d\theta}{d\varepsilon} = \frac{n\cdot\cos(\frac{\pi}{6}+\varepsilon)}{\sqrt{1-n^2\sin^2(\frac{\pi}{6}+\varepsilon)}} - \frac{n\cos(\frac{\pi}{6}-\varepsilon)}{\sqrt{1-n^2\sin^2(\frac{\pi}{6}-\varepsilon)}} = 0$

$\varepsilon = 0$. $\frac{d\theta}{d\varepsilon}=0$ ✓

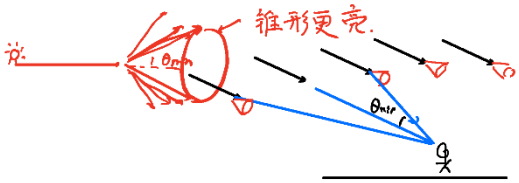
$\theta_{min} = 0.378$ rad $= 21.5°$. → 即为晕.



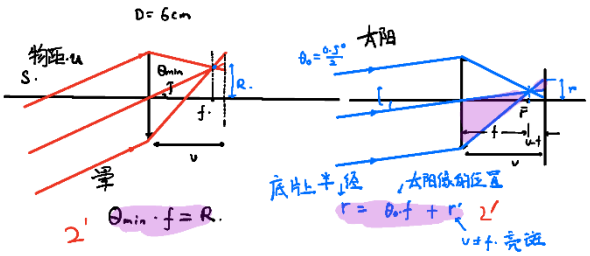
$\hat{n}$ 任意 在θ_{min}附近时.

有一群相邻$\hat{n}$对应同样θ.

⇒ 显著亮.



锥形更亮.



底片上半径 太阳成像位置

$r = \theta_0 f + r'$ 2'

$v \neq f$. 亮斑

相似△. $\frac{\frac{D}{2}}{f} = \frac{r'}{v-f}$ 2'

$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$. ⇒ $v = \frac{fu}{u-f}$

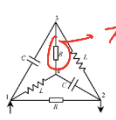
$v - f = \frac{fu}{u-f} - f = \frac{f^2}{u-f} \simeq \frac{f^2}{u}$.

$r = \theta_0 f + \frac{D}{2f}\cdot\frac{f^2}{u} = f\cdot(\theta_0 + \frac{D}{2u})$.

$\frac{R}{r} = 15 = \frac{\theta_{min}}{\theta_0 + \frac{D}{2u}}$ ⇒ $u = 1.4$m 2'

5. (40分)

如图所示一个电路图，是正四面体的几个棱边组成的。电阻为R，电感为L，电容为C加载的电压是$V = V_0\cos\omega t$，电阻满足$R = \sqrt{\frac{L}{C}}$，且有$\omega = \frac{1}{\sqrt{LC}}$。求出电路的总电流。



不可电桥

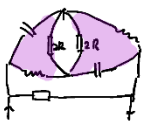
$Z_C = \frac{1}{i\omega C} = \frac{\sqrt{LC}}{i} = -iR.$

$Z_L = i\omega L = iR.$

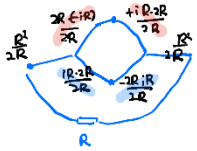
法0. 基xx 3个图10' 结果0'

法1. Δ-Y 对称变.

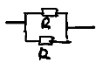
分母≠0.



$R \Rightarrow \frac{R}{2}$

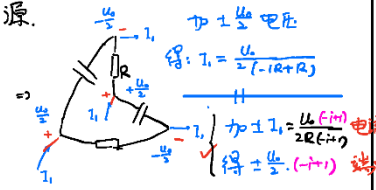


$R_1 = \frac{R_{12}\cdot R_{13}}{R_{12}+R_{23}+R_{31}} = \frac{1}{2}R.$

$\Rightarrow$  $\Rightarrow I = \frac{U_0}{R}$; $I(t) = \frac{2V_0}{R}\cos\omega t$ 4o'

法2. 叠加电流源.



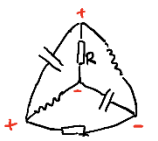
$\Rightarrow$ 

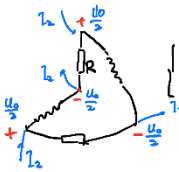
加 $\pm\frac{U_0}{2}$ 电压

得: $I_1 = \frac{U_0}{2(-iR+R)}$

加 $\pm I_1 = \frac{U_0(i+1)}{2R(-i+1)}$ 电流源

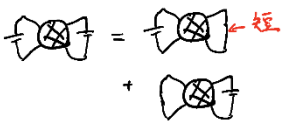
得 $\pm\frac{U_0}{2}\cdot(-i+1)$ 两个电压

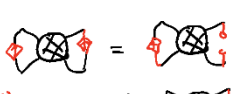


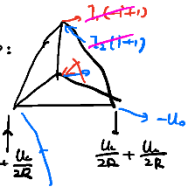


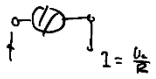
$I_2 = \frac{U_0(i+1)}{2(iR+R)}$

$U = \pm\frac{U_0}{2}(1+i)$

电压加 ✓ 

电流源叠加. ✓ 

再加: 

$\Rightarrow$ 

$I = \frac{U_0}{R}$

$I: \frac{U_0}{2R} + \frac{U_0}{2R}$

$\Delta U = U_0 - (-U_0) = 2U_0$

$U = \frac{U_0}{2}(1-i) + \frac{U_0}{2}(1+i) = U_0$

$\Rightarrow I = \frac{2U_0}{R}$ ← 4o'

6. (40分)

波尔（Bohr）和萎了(Wheeler)曾用液体模型来分析原子核的情况。并且在1939年算出了某个裂变有关的极限。我们现在就来重复一下这个工作。我们用Z来表示核子中的质子数。A来表示原子中的核子数。

考虑液滴模型的话，能量就可以表示成表面能和电能两个部分。我们用U_A来表示表面能，用U_E表示电能。显然表面能和半径的平方成正比，也就是$U_A = a_S R^2$其中a_S是一个系数。没有量纲。R是原子核的半径。我们不妨假设原子核为球……，要不太难算了。(大神可以尝试计算椭球) 这样的话$R = r_0 A^{1/3}$其中$r_0 = 1.2fm(1fm = 10^{-15}m)$如图所示，考虑一个半径为$R_0$的母核裂变为两个半径分别为$R_1$ R_2的子核。（原文是女核daughter nuclei，这样就不性别歧视了!)我们假设1, 2两个核的质量之比为f。母核和子核的质量密度和电荷密度都保持不变。



$\frac{A\cdot m_p}{\frac{4}{3}\pi r_0^3\cdot A} = \rho = 2.31\times10^{17}kg/m^3$ 8'

a)估算原子核的质量密度，假设质子和中子的质量基本一样。

b)两个子核的表面能之和可以写$U_A' = a_S R_0^2\alpha$也就是说是原来母核的表面能的一个系数倍，请求出这个系数α用f表示。

c)求出母核的静电能。U_E'用MeV作为单位，以及用Z, A啥的表示。

d)如果是刚刚裂变，也就是三个图中中间那个图的情况，我们用Z_1, A_1, Z_2, A_2表示两个子核的质子数和核子数。请求出两个子核的总静电能U_E'

e)和刚才思路类似，子核静电能应该是母核静电能的某个系数倍，求出这个系数β

f)*（后面几问较难，我认为复赛模拟时候不计分，*however*，今年湖北，加油）真正的裂变不是第一个图到第二个图的变化，而是第一个图到第三个图的变化，也就是两个子核的距离可以认为无限远。如果在一个裂变的过程中释放的能量用Q(单位使用为MeV) 表示A，Z分别表示母核的核子数和质子数。请求出常数a_S用Q, Z, A, α, γ等表示。其中γ是本小问计算过程中可能出现的只和f有关的常数。

g) *若知道核反应$^1_0n + ^{235}_{92}U \to ^{141}_{56}Ba + ^{92}_{36}Kr + 3(^1_0n)$中$Q = 173.2$MeV,求出$a_S$(单位MeV/$fm^{-2}$)

h) *裂变能发生的条件就是$\frac{Z^2}{C} > C$这里面C就是一个可以用前面计算过的a_S和f表示的量了。求出C

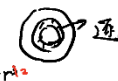
i)求出极限，也就是C的最小值。仔细考虑f的物理意义，以及极限情况下它的取值。

(2) $A_1 + A_2 = A$. $A_1 = f\cdot A_2$

$U_E = \alpha\cdot a_S R_0^2 = a_S R_1^2 + a_S R_2^2$

$R_1 = R_0\cdot(\frac{f}{1+f})^{\frac{1}{3}}$ $R_2 = R_0(\frac{1}{1+f})^{\frac{1}{3}}$

$\alpha = \frac{1+f^{2/3}}{(1+f)^{2/3}}$ 8'

(3). $r\sim r+dr$ 抽走  逐层抽走

$U(r) = \frac{\rho\cdot\frac{4}{3}\pi r^3\cdot z}{4\pi\varepsilon_0\cdot r}$

$W = U(r)\cdot 4\pi r^2 dr\cdot\rho$

$= \frac{\rho^2}{4\pi\varepsilon_0}\cdot\frac{4}{3}\pi\cdot 4\pi\int_0^{r_0} r^4 dr$ $\to z^2e^2$

$= \frac{\rho^2}{4\pi\varepsilon_0}\cdot\frac{4}{3}\pi\cdot\frac{4\pi}{3}\cdot\frac{3}{5}\cdot\frac{r_0^5}{r_0}$

$= \frac{3}{5}\cdot\frac{1}{4\pi\varepsilon_0}\cdot\frac{z^2e^2}{r_0}$ 8'

(4).



$U = \frac{3}{5}\cdot\frac{1}{4\pi\varepsilon}\frac{z_1^2e^2}{R_1} + \frac{3}{5}\frac{1}{4\pi\varepsilon}\frac{z_2^2e^2}{R_2} + \frac{z_1 z_2 e^2}{4\pi\varepsilon_0(R_1+R_2)}$

$R_1 = (\frac{f}{1+f}A_0)^{1/3}\cdot r_0$ $R_2 = (\frac{1}{1+f}A_0)^{1/3} r_0$.

$z_1^2 = z^2(\frac{f}{1+f})^2$ $z_2^2 = z^2(\frac{1}{1+f})^2$

$= \frac{3}{5}\frac{1}{4\pi\varepsilon_0}\cdot\frac{z^2e^2}{r_0A^{1/3}}\cdot(\frac{f}{1+f})^{\frac{2}{3}} + \frac{3}{5}\frac{1}{4\pi\varepsilon_0}\frac{z^2e^2}{r_0A^{1/3}}\cdot(\frac{1}{1+f})^{\frac{2}{3}}$

$+\cdot\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_0A^{1/3}}\cdot\frac{(\frac{f}{1+f}+\frac{1}{1+f})^2}{(\frac{f}{1+f})^{1/3}+(\frac{1}{1+f})^{1/3}}$ 4'

$= \frac{1}{4\pi\varepsilon_0}\frac{3}{5}\frac{z^2e^2}{r_0A^{1/3}}\cdot[(\frac{f}{1+f})^{2/3} + (\frac{1}{1+f})^{2/3} + \frac{f}{(1+f^{1/3})\cdot(1+f)^{2/3}}]$

(f). $\Rightarrow$ $\beta = [(\frac{f}{1+f})^{2/3} + (\frac{1}{1+f})^{2/3} + \frac{f}{(1+f^{1/3})\cdot(1+f)^{2/3}}]$ 4'

(6). $a_s = \dfrac{Q-(1-\gamma)\cdot\frac{3e^2Z^2}{5\cdot 4\pi\varepsilon_0 r_0 A_0^{1/3}}}{(1-\alpha)r_0^2 A_0^{2/3}}$

$\gamma = \dfrac{1+f^{5/3}}{(1+f^{5/3})}$.

(7). $a_0 = 12.7\ MeV/(fm)^2$

(8) $Q=0$. $c = \dfrac{a_s r_0^2(\alpha-1)}{a_{72}(1-\gamma)}$

(9) $f=1$. $c = 17.8$.

不是考题

7. (40分)

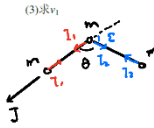
一根长度为l的柔软的不可伸长的无弹性均匀细绳，质量为m，摆在光滑水平面上，形成一个角度为$\theta < 2\pi$的圆弧。用一个冲量I着切线方向向外冲击绳子一端，如左图所示，求很短之间之内，绳子另一端的速度v_1为多少。



(1)为了求解这个问题，我们先做一个简单问题，如中间的图所示，3个质量为m的球，被两根绳子连接，夹角为θ，用冲量I冲击第一个球时，求第三个球速度v_3

(2)复杂一点，$N+1$个球，用N段绳连接，其他条件一样，求最后一个球速度v_3

(3)求v_1



(1). $\xi = \pi - \theta$. $\cos\xi = -\cos\theta$.

4' 1: $\dfrac{I-I_1}{m} = \dfrac{I_1}{m} + \dfrac{I_2}{m}\cdot\cos\theta$. 与它U一样.

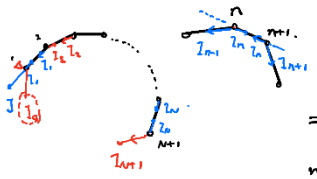
4' 2: $-\dfrac{I_1}{m}\cos\theta - \dfrac{I_2}{m} = \dfrac{I_2}{m}$. ②

$\dfrac{I_2}{m} = -\dfrac{1}{2}\cdot\dfrac{I_1}{m}\cos\theta$

$\dfrac{I}{m} = \dfrac{2I_1}{m} - \dfrac{I_1}{2m}\cos^2\theta$;

$I_1 = \dfrac{I\cdot 2}{4-\cos^2\theta}$; $I_2 = -\dfrac{I\cos\theta}{4-\cos^2\theta}$.

$\Rightarrow v_2 = -\dfrac{I}{m}\dfrac{\cos\theta}{4-\cos^2\theta}$. 8'



第n段: $\dfrac{n\text{-}1}{m}$

$-\dfrac{I_{n-1}\cos\theta}{m} - \dfrac{I_n}{m} = \dfrac{I_n}{m} + \dfrac{I_{n+1}}{m}\cdot\cos\theta$

$\Rightarrow I_{n-1}\cos\theta + 2I_n + I_{n+1}\cdot\cos\theta = 0$. 4'

$n=1$ ↘ 令 $-I_0\cos\theta = I$. 2' $I_{N+1} = 0$ 2' ↙ $n=N$

用I_0模拟I. 用I_{N+1}模拟末尾.

代入 $I_n = A\cdot\beta^n$ 代入

$\beta^2 + \dfrac{2\beta}{\cos\theta} + 1 = 0 \Rightarrow \beta = \dfrac{-\frac{2}{\cos\theta} \pm \sqrt{\frac{4}{\cos^2\theta}-4}}{2}$

$= -\dfrac{1}{\cos\theta} \pm \dfrac{\sqrt{1-\cos^2\theta}}{\cos\theta}$

$\beta_2 = -\dfrac{1\pm\sin\theta}{\cos\theta}$ 4'

$\beta_+ > 1 > \beta_- > 0$. $\beta_+\cdot\beta_- = 1$. $\beta_+ + \beta_- = -\dfrac{1}{\cos\theta} > 0$.

$I_n = A\beta_+^n + B\beta_-^n$.

$I_n = A\beta_+^n + B\beta_-^n$

$-I_0\cos\theta = I$. $I_{N+1} = 0$

$\Rightarrow A+B = -\dfrac{I}{\cos\theta}$. 2'

$A\beta_+^{N+1} + B\beta_-^{N+1} = 0$ 2'; $B = -A\cdot\left(\dfrac{\beta_+}{\beta_-}\right)^{N+1}$

$A = \dfrac{-\frac{I}{\cos\theta}}{1-\left(\frac{\beta_+}{\beta_-}\right)^{N+1}}$; $B = \dfrac{-\frac{I}{\cos\theta}}{1-\left(\frac{\beta_-}{\beta_+}\right)^{N+1}}$.

$\Rightarrow I_n = \dfrac{-\frac{I}{\cos\theta}\cdot\beta_+^n}{1-\left(\frac{\beta_+}{\beta_-}\right)^{N+1}} + \dfrac{-\frac{I}{\cos\theta}\beta_-^n}{1-\left(\frac{\beta_-}{\beta_+}\right)^{N+1}}$ 4'

最后. $\dfrac{I_N}{m}$.

$\theta = \pi - \dfrac{\pi}{N}$; $N\to\infty$

$\lim_{x\to\infty}\left(1+x\right)^{\frac{1}{x}} = e$. 秒入.

$v_{末} = \lim_{N\to\infty}\dfrac{I_n}{m} = \dfrac{I}{m}\dfrac{2\theta e^\theta}{e^{2\theta}-1}$ 4'

用3查 $\vec{U}(s)$ 关联.

8. (40分)

考虑一些几乎没有相互作用的静质量为m_0的质点构成的理想气体，数密度为n。其在动量空间的概率密度为$dP = f(p_x,p_y,p_z)dp_xdp_ydp_z$，在非相对论情况下，$f(p_x,p_y,p_z) = 4\pi p^2(\frac{1}{2\pi m_0 kT})^{3/2}e^{-\frac{p^2}{2m_0kT}}$。

(1)在非相对论场景下，计算温度为T时体系的压强。

(2)当温度很高的时候，我们需要考虑侯义相对论修正，取$\varepsilon = \frac{kT}{m_0c^2}$为一阶小量，假设分布律仍然可以写为$f(p_x,p_y,p_z) = 4\pi p^2 Ae^{-\frac{E}{kT}}$，其中$E = \sqrt{m_0^2c^4+p^2c^2}$。保留到$\varepsilon$的一阶小量，计算$A$，以及平均能量$\overline{E}$，计算平均压强。

积分提示：

令$G_n(\alpha) = \int_0^\infty x^n e^{-\alpha x^2}dx$，给出以下积分结果

$G_0(\alpha) = \frac{1}{2}\sqrt{\frac{\pi}{\alpha}}, G_1(\alpha) = \frac{1}{2\alpha}, G_2(\alpha) = \frac{1}{4}\sqrt{\frac{\pi}{\alpha^3}}, G_3(\alpha) = \frac{1}{2\alpha^2}, G_4(\alpha) = \frac{3}{8}\sqrt{\frac{\pi}{\alpha^5}}, G_5(\alpha) = \frac{1}{\alpha^3}$,

$G_6(\alpha) = \frac{15}{16}\sqrt{\frac{\pi}{\alpha^7}}, G_7(\alpha) = \frac{3}{\alpha^4}, G_8(\alpha) = \frac{105}{32}\sqrt{\frac{\pi}{\alpha^9}}$

(1) $f(p) = 4\pi p^2 \cdot (\frac{1}{2\pi m_0 kT})^{3/2} e^{-\frac{p^2}{2m_0kT}}$ $= (\frac{m_0}{2\pi kT})^{3/2} e^{-\frac{\frac{1}{2}m_0v^2}{kT}}$ $f(v)$

x方向 ds, $v_x dt$

$p_x \sim p_x + dp_x$.

$P_{压} = \frac{\int dS \cdot n \cdot \frac{p_x}{m} \cdot dt \cdot 2p_x \cdot f(p_x)\,dp_x}{dS \cdot dt}$ ∫

$f(v) = 4\pi v^2 \cdot C\,e^{-\frac{mv^2}{2kT}}$

类比.

$f(p_x) = (\frac{1}{2\pi m_0 kT})^{1/2} e^{-\frac{p_x^2}{2m_0kT}}$ ∫'

$P_{压} = \int_0^\infty dp_x \cdot n \cdot 2 \cdot \frac{p_x^2}{m} \cdot (\frac{1}{2\pi m_0 kT})^{1/2} e^{-\frac{p_x^2}{2m_0kT}}$

$u \cdot \sqrt{\quad}$ $u^2 \cdot (\quad)$

$\frac{p_x}{\sqrt{2m_0kT}} = u$.

$P_{压} = \frac{2n}{m} \cdot (\frac{1}{2}m_0kT)^{3/2} \cdot (\frac{1}{2\pi m_0 kT})^{1/2} \cdot \int_0^\infty u^2 e^{-u^2}\,du$ ∫'

$\frac{1}{4}\cdot\sqrt{\pi}$

$= \frac{n}{m} \cdot mkT = nkT$. ←∫'

(2) $f(p) = 4\pi p^2 \cdot A e^{-E/kT}$.

$\varepsilon = \frac{m_0c^2}{kT}$ 为一阶. ← 不改 $kT \gg m_0c^2$.

粒子近光速. pc. 大.

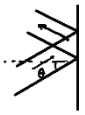
$E = \sqrt{m_0^2c^4+p^2c^2}$

$= pc \cdot (1 + (\frac{m_0c^2}{pc})^2)^{1/2}$

$\simeq pc\,(1 + \frac{1}{2}(\frac{m_0c^2}{pc})^2)$. 近大

$f(p) = 4\pi p^2 \cdot A\ e^{-\frac{1}{kT}\cdot(pc + \frac{(m_0c^2)^2}{2pc})}$.

压强 θ方向, 立体角 $d\Omega$.



$n \cdot \frac{d\Omega}{4\pi} \cdot ds \cdot c\cos\theta\,dt \cdot 2p\cos\theta \cdot f(p)$.

角概率.

$P_{压} = \int_{\frac{\pi}{2}}^{0}\int_0^\infty n \cdot \frac{1}{4\pi} \cdot c\cos\theta \cdot 2p\cos\theta \cdot 4\pi p^2 \cdot A\ e^{-\frac{pc}{kT}}\,dp\,d\Omega$

$\int_0^\infty 4\pi p^2 \cdot A\ e^{-\frac{1}{kT}\cdot pc}\,dp = 1$. ← 归一.

$P_{压} = \frac{1}{2}nkT$.

改:

$\varepsilon = \frac{kT}{m_0c^2}$ 为一阶 $pc \ll m_0c^2$

$\sqrt{m_0^2c^4+p^2c^2} = m_0c^2 \cdot (1 + \frac{p^2c^2}{m_0^2c^4})^{1/2}$ $\frac{n(n-1)}{2}$.

$\simeq m_0c^2\,(1 + \frac{1}{2}\frac{p^2c^2}{m_0^2c^4} - \frac{1}{8}\frac{p^4c^4}{m_0^4c^8})$

$1 = \int f(p)\,dp = \int A \cdot 4\pi p^2 \cdot e^{-\frac{m_0c^2}{kT} - \frac{1}{2}\frac{p^2c^2}{m_0c^2}\frac{1}{kT} - \frac{1}{8}\frac{p^4c^4}{m_0^4c^8}\frac{1}{kT}}\,dp$

$Ae^{-\frac{m_0c^2}{kT}}$ ~ 原归一.

多. $\frac{p^2c^2}{m_0^2c^4}$ 为 $O(\varepsilon)$.

$A \cdot e^{-\frac{m_0c^2}{kT}} \cdot \int_0^\infty 4\pi p^2 \cdot (1 - \frac{1}{8}\varepsilon u^4) \cdot e^{-u^2}\,du = 1$.

$\frac{p^2}{2m_0kT} = u^2$. $\frac{1}{8}\frac{p^4c^4}{m_0^2c^4}\frac{1}{kT} = \frac{1}{8}\cdot u^4 \cdot \frac{kT}{m_0c^2} := \frac{1}{8}u^4 \cdot \varepsilon$.

$e^{-u^2 - \frac{1}{8}\varepsilon u^4} = e^{-u^2}(1 - \frac{1}{8}\varepsilon u^4)$. ε 一阶小.

$O(\varepsilon^0)$: $\int_0^\infty u^2 e^{-u^2}\,du \to \frac{\sqrt{\pi}}{4}$

$O(\varepsilon)$: $\int_0^\infty -\frac{1}{8}\varepsilon u^4 e^{-u^2}\,du \to \frac{3}{8}\sqrt{\pi}(-\frac{1}{8})\varepsilon$ 相比. $-\frac{3}{2}\cdot\frac{1}{8}\varepsilon$.

$Ae^{-\frac{m_0c^2}{kT}}(1 - \frac{3}{16}\varepsilon)$. = 原 $A = (\frac{1}{2\pi m_0kT})^{3/2}$.

$\Rightarrow Ae^{-\frac{m_0c^2}{kT}} = (1 + \frac{3}{16}\varepsilon)\cdot(\frac{1}{2\pi m_0kT})^{3/2}$.

压强: $P_{压} = \int_0^\infty\int_{\frac{\pi}{2}} \frac{d\Omega}{4\pi} \cdot n \cdot v_x \cdot p\cos\theta \cdot 2 \cdot f(p)\,dp$

$v\cos\theta$

$p = \frac{m_0v/c}{\sqrt{1-v^2/c^2}} \Rightarrow \frac{p}{m_0c} = \frac{v/c}{\sqrt{1-v^2/c^2}}$

$\Rightarrow \frac{p^2}{m_0^2c^2} = \frac{v^2/c^2}{1-v^2/c^2} \Rightarrow 1 + \frac{p^2}{m_0^2c^2} = \frac{1}{1-v^2/c^2}$

$\Rightarrow 1 - \frac{v^2}{c^2} = \frac{1}{1+\frac{p^2}{m_0^2c^2}}$ $\frac{v^2}{c^2} = 1 - \frac{1}{1+p^2/m_0^2c^2}$

$\frac{v}{c} = \frac{p/m_0c}{\sqrt{1+p^2/m_0^2c^2}}$ $\Rightarrow v \simeq \frac{p}{m_0}\cdot(1 - \frac{1}{2}\cdot\frac{p^2}{m_0^2c^2})$.

$P_{压} = \int_0^\infty\int_{\frac{\pi}{2}} \frac{d\Omega}{4\pi} \cdot n \cdot \frac{p^2}{m_0} 2\cos^2\theta \cdot (1 - \frac{1}{2}\frac{p^2}{m_0^2c^2}) \cdot f(p)\,dp$. εu^2

原相同 $\frac{p^2}{2m_0kT} = u^2$

原 $\cdot(1 - \varepsilon u^2)$. $-\frac{p^2}{2m_0kT}\cdot\frac{kT}{m_0c^2} \to \varepsilon$.

原 $\cdot \int u^2 e^{-u^2}\,du \to \int u^2(1-\varepsilon u^4)\,du$

$P_{压} = P_{压原}\frac{(1 - \varepsilon\frac{3}{2})}{(1 - \varepsilon\frac{3}{2}\cdot\frac{1}{5})} = P_{压原}\cdot(1 - \varepsilon\frac{3}{2}\cdot\frac{7}{8})$.

$= nkT \cdot (1 - \frac{21}{16}\cdot\frac{kT}{m_0c^2})$. ← $O(\varepsilon)$.